CO1 - AT-1
Short Answers

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1. Vehicle Management system.

Introduction

An EER (Enhanced entity-Relationship) Diagram Extends the basic ER model with additional concepts such as specialization, inheritance and constraints which are needed when entities naturally divide into subtypes with their own distinct attributes

i) Supertype and Subtypes.

• Supertype: Vehicle - the generic entity representing any Vehicle in the company's fleet, holding attributes common to every vehicle regardless of type.

• Subtype: Car and Truck - specialized entities that represent two distinct more specific categories of vehicle, Each

ii) Common Attributes of vehicles

Attributes	Description
Vehicle_ID(PK)	Uniquely identifies every vehicals
Registration_number	Government issued registration
Manufacturer	Company that build the vehicle
model	specific model name
year	manufacturing / purchase year

iii) Specialized Attributes

Car- specific attributes (apply only to cars, not Trucks

* Number_of Doors – relebant to passenger confort
* fuel_type - petrol, Diesel, Electric
* Seating_capacity - number of passengers the car can carry

Truck – specific attributes (apply only to trucks not cars.

* Load – capacity - maximum weight the truck can carry
* Truck - Type - Eg - flated container tanke

iv) Primary key & inheritance

Vehicle I is declared as the primary key of the vehicle entity. Because car and truck are specializations of vehicle

v) ISA Relationship

The relationship between vehicle and its subtypes is represented using the ISA (IS-A) notation read as car IS-A vehicle and "Truck IS-A Vehicle" in standard EER notation this is drawn as a triangle

vi) Disjoint / overlapping constraint.

A given vehicle entity instance can belong to only one subtype at a time it is either a car or a truck never both simultaneously.

Bank Management System ER-Diagram

i) Major Entities

Customer, Account, Branch, Employee, Transaction, Loan, Loan-payment

Each entity represents a distinct real world object that the bank needs to track independently with its own attributes and identity

ii) Attributes

iii) Primary Key

Entity	Attributes	Primary Key
Customer	Customer_ID, Name address, Phone	Customer_ID
Account	Account_No, Account_Type Balance opening_Date	Account_No
Branch	Branch_ID, Branch_name, Location IFSC_code	Branch_ID
Employee	Transcaction_ID, Transaction_Date Transaction_Type, Account	Employee_ID
Loan_Payment	Payment_ID, Payment_Date, Payment Account	Payment_ID

iv) Relationships and cardinalities

Entities	Relationship	Cardinality
Customer Account	Holds	1: N (one customer can hold many accounts
Branch - Account	MAINTAINS	1:N (one branch maintains many accounts.
Branch - employee	EMPLOYE	1:N
Repaid_By	Loan_Loan_Payment	1:N (one loan - has many payments.

vi)

Customer	Account	Branch
Customer_ID	Account_No	Branch_ID
Customer_Name	Account_Type	Branch_Name

Address

Loan

Loan_ID

Lonan_Type

E-Commerce website - EER Diagrams

i) & ii)

Entities, Attributes, Primary key

Entities	Attributes	Primary key
Customer	Customer_ID, Name, Email, phone, Address	Customer_ID
Product	Product_ID, Product_Name Price, Stock_Quality, Description	Product_ID
Category	Category_ID; Category_Name	Category_ID
Order	Order_ID, Order_Date, Order_Status	Order_ID
Payment	Payment_ID, Payment_Data, Payment Status	Payment_ID
Delivery	Delivery_ID, Deliver_Date, Delivery Status	Delivery_ID

iii) Order_product relationship

A many-to many relationship contains between order and product resolved via an associative entity order with attributes Quality and unite_price.

4&5 Specialization of product.

product (superstype) → Electronics, clothing, grocery (subtypes)

* Electronics. Warranty_period, Brand
* Clothing - Size, color, Material
* Grocery - Expiry_Date, weight

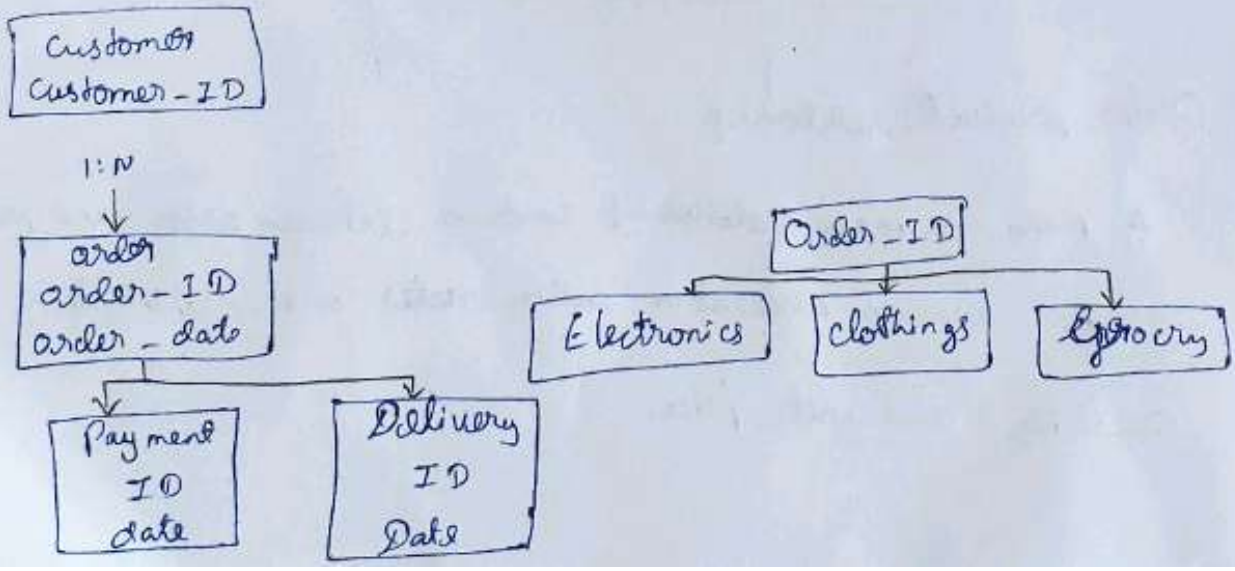
6) Disjoint / Overlapping & Total / partial.

Disjoint, Total - each product belongs to exactly one category every product must be classified

7+8 Relationships and cardinalities

Relationship	Cardinality
Customer_order (Places)	1:N
Order_payment (HAS)	1:1
Order - Delivery (HAS)	1:1
Order - product (via order_Item	M:N

9)



Hospital Management System

1) Major Entities

The major entities are staff ; Doctor, nurse, admin, Department Appointment 2-4 staff specialisation and attributes.

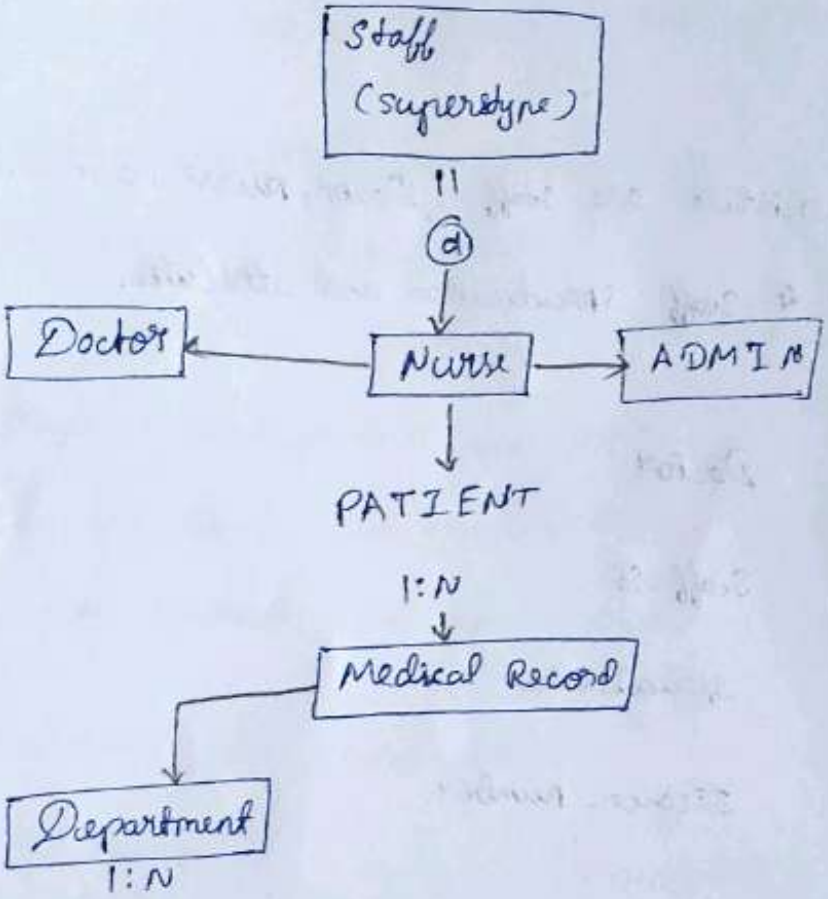
Staff	Doctor
Staff_id	Staff_id
Name	Specialisation
Address	Licence_number
Phone	
Salary	

5-7 is Week entity and Identifying relation medical record is cock entity.

Partial key - Record - Rate.

Identity realionship - HAS - record.

EER Diagram


Staff
(superstype)
d
Doctor
Nurse
ADMIN
PATIENT
1:N
Medical Record
Department
1:N

Database Administration

1) A Database administerator is responsible for installing configuring, securing, maintaing and mainitoring an organisation database system.

2) The DBA helps design database schemas table relationship indexs constraints DBA selects suitable database software configurase it when required

3) The DBA protects data from all the unized access by implementing authentication authorization encryption and auditing

4) The DBA schedule and verifies redes database backups, Recovery procedureled are and tested to restores data after a accidental deletion.

5) The DBA monitors query execution time usuage memory consumption dest activity and database day stat.

6) The DBA maintains data accuracy and consistency by enforcing primary has entrands is consistance values related rules

7) DBA manages data base storage capacity transaction key UD grading software maintain disk space

8) DBA identies and resolves data base error connection failure, deallocks corruption perpormens problem

9) Use and privi edge management DBA Executes modifies diable database user accounts.

10) The DBA ensure database available scal ever mechanism where appropriate